

Conduction, Convection and Radiation: Comparing Heat Transfer

SimuPhysics Heat Transfer Simulation · SimuPhysics

Years 8-9	~30 minutes	Science · Physics
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Simulation: <https://simuphysics.com/view/thermodynamics-heat-transfer-simulation.html>

Curriculum links: AC9S8U05 · AC9S9U05 · thermal physics

Learning intentions

- I can describe the differences between conduction, convection and radiation.
- I can identify which variables affect the rate of heat transfer in each mechanism.
- I can use the law of conservation of energy to explain where thermal energy goes during a transfer.

Getting started

Open the SimuPhysics Heat Transfer Simulation. Start with the conduction setting, choose a material, and set a temperature difference between the two ends or objects before you press start.

Tasks

1. Set up conduction between two objects with a large temperature difference. Describe what happens to the temperature of each object over time.

2. Change the material or increase the temperature difference. Describe how this affects the speed of heat transfer.

3. Test conduction with three different material or temperature-difference settings. Record the approximate time taken to reach a similar temperature change each time.

Material/setting	Time for noticeable heat transfer

4. Using your table, explain what factors seem to speed up or slow down conduction.

5. Switch to the convection setting. Describe what you observe happening to the heated fluid (air or liquid) as it warms and cools.

6. Switch to the radiation setting. Unlike conduction and convection, radiation doesn't need particles to touch or flow between objects. Describe one difference you notice about how radiation transfers heat compared to the other two mechanisms.

7. Using the law of conservation of energy, explain what happens to the thermal energy that leaves the hotter object in each of the three mechanisms — where does it go?

Extension (optional)

A thermos flask is designed to slow down all three types of heat transfer at once. Using what you've learned from the simulation, suggest one design feature that would reduce conduction, one that would reduce convection, and one that would reduce radiation.
